AS · Unit 2 · WPH12 Waves & Electricity

Refraction, Diffraction & Polarisation

Light bends when it crosses into a new material, spreads out when it squeezes through a gap, and can be filtered by the direction it vibrates in. All three behaviours come from picturing light as wavefronts that slow down, spread out and line up. On the way you'll derive Snell's law and the grating equation, and then use a laser and a grating to measure the wavelength of light (Core Practical 6).

Refraction and refractive index

A straw in a glass of water looks bent at the surface, and a swimming pool always looks shallower than it is. Both happen because light travels at different speeds in different materials: 3.00×10^8 m s^{-1} in a vacuum, but only about 2.0×10^8 m s^{-1} in glass. When a wave changes speed at a boundary at an angle, it changes direction. This is **refraction**.

Why a change of speed means a change of direction

Picture a row of people marching arm in arm from a road onto sand, crossing the edge at an angle. The end that reaches the sand first slows down while the rest keep going, so the row swings round. A wavefront does the same: the part that enters the slower medium first falls behind, the wavefront pivots, and the direction of travel (always perpendicular to the wavefronts) turns **towards the normal**. A wave arriving along the normal slows all at once and doesn't turn.

The **frequency does not change**. Every crest that arrives at the boundary makes exactly one crest on the other side, so the number of waves per second is fixed by the source. Because $v = f\lambda$, when the speed drops the **wavelength shrinks by the same factor**.

DEFINITION

The **refractive index** n of a material is the speed of light in a vacuum divided by the speed of light in the material: $n = c/v$. It has no units and is never less than 1. For air $n = 1.0003$, which you can take as 1.00.

Refractive index FORMULA SHEET

$$n = \frac{c}{v}$$

c: speed of light in a vacuum / m s^{-1}
v: speed of light in the material / m s^{-1}

Snell's law FORMULA SHEET

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

θ_1: angle of incidence θ_2: angle of refraction
both measured from the normal

Material	Refractive index n	Speed of light / m s^{-1}	Critical angle (to air)
vacuum	1 (exactly)	3.00×10^8	none
air	1.00	3.00×10^8	none
water	1.33	2.26×10^8	48.8°
glass (typical)	1.50	2.00×10^8	41.8°
diamond	2.42	1.24×10^8	24.4°

Snell's law says the product $n \sin\theta$ is the same on both sides of the boundary. So on the side with the **bigger** n, $\sin\theta$ must be **smaller**: the ray is closer to the normal in the slower material. Going into a slower medium the ray bends **towards** the normal; going into a faster one it bends **away** from it.

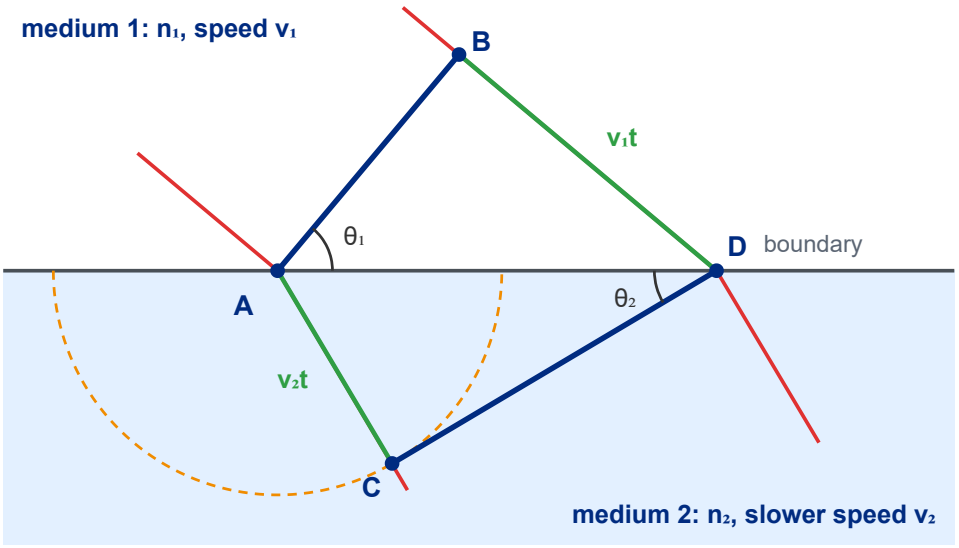
COMMON MISTAKE

All the angles in Snell's law are measured from the **normal** (the line at 90° to the surface), not from the surface itself. If a question gives you the angle between the ray and the surface, subtract it from 90° first.

Drag on the top half of the diagram to aim the ray box, or use the slider. The faint ray going back up is the partly reflected ray.

Deriving Snell's law from Huygens' wavefronts

Huygens' idea (more on it in the diffraction section) is that every point on a wavefront acts as a source of small circular wavelets, and the new wavefront is the line that touches all of them. Let's use it on a wavefront arriving at a boundary.



Wavefront AB (navy) reaches the boundary at A first. While B travels $v_1 t$ to D, the wavelet from A grows to radius $v_2 t$. The new wavefront CD touches that wavelet. Red lines are rays, at 90° to the wavefronts.

Snell's law from Huygens' construction

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A plane wavefront AB meets the boundary at an angle, so end A touches it first. The angle between the wavefront and the boundary equals the angle of incidence θ_1.

   The wavefront is at 90° to the ray and the boundary is at 90° to the normal, so turning both lines through 90° keeps the angle between them the same.

2. In a time t, end B (still in medium 1) travels to D at speed v_1. Meanwhile the wavelet from A spreads into medium 2 at the slower speed v_2:

$$BD = v_1 t, \qquad AC = v_2 t$$

3. The new wavefront is the line from D that just touches the wavelet, CD. A tangent meets a radius at 90°, so AC is perpendicular to CD, and AC is the direction of the refracted ray. The angle at D between the boundary and the new wavefront is the angle of refraction θ_2, for the same reason as in step 1.

4 Triangles ABD and ACD are both right-angled and share the hypotenuse AD:

$$\sin\theta_1 = \frac{BD}{AD} = \frac{v_1 t}{AD}, \qquad \sin\theta_2 = \frac{AC}{AD} = \frac{v_2 t}{AD}$$

5 Divide one by the other. Both t and AD cancel:

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2}$$

Since f is the same on both sides, this ratio also equals λ_1/λ_2.

6 Now use $v = c/n$, so $v_1/v_2 = (c/n_1)/(c/n_2) = n_2/n_1$:

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_2}{n_1} \quad \Rightarrow \quad \boxed{n_1 \sin\theta_1 = n_2 \sin\theta_2}$$

WORKED EXAMPLE

Light of wavelength 600 nm in air enters a glass block ($n = 1.50$) at an angle of incidence of 40°. Calculate (a) the angle of refraction, (b) the speed of light in the glass, (c) the wavelength and frequency of the light in the glass.

Show solution

(a) Snell's law with $n_1 = 1.00$:

$$\sin\theta_2 = \frac{n_1 \sin\theta_1}{n_2} = \frac{1.00 \times \sin 40^\circ}{1.50} = \frac{0.643}{1.50} = 0.429 \quad \Rightarrow \quad \theta_2 = 25.4^\circ$$

(b) $v = \dfrac{c}{n} = \dfrac{3.00 \times 10^8}{1.50} = 2.00 \times 10^8 \text{ m s}^{-1}$

(c) The frequency is unchanged, so the wavelength falls by the same factor as the speed:

$$f = \frac{c}{\lambda_{\text{air}}} = \frac{3.00 \times 10^8}{600 \times 10^{-9}} = 5.00 \times 10^{14} \text{ Hz}, \qquad \lambda_{\text{glass}} = \frac{v}{f} = \frac{2.00 \times 10^8}{5.00 \times 10^{14}} = 4.00 \times 10^{-7} \text{ m} = 400 \text{ nm}$$

Reflection, transmission and total internal reflection

Whenever a wave reaches a boundary between two media, **some of it is transmitted** (and refracted) and **some of it is reflected**, which is why a window at night shows your reflection while still letting light through. The reflected part obeys angle of reflection = angle of incidence, and the more different the two media, the bigger the reflected share. Sound, ultrasound and radio behave the same way; ultrasound scanning relies on the small echo from every tissue boundary (pulse-echo, on the *Progressive Waves* page).

The critical angle

Now send light the other way, from a **higher** refractive index into a **lower** one (glass into air, say). The ray speeds up and bends **away** from the normal, so θ_2 is always bigger than θ_1. Turn up θ_1 and θ_2 reaches 90° first: the refracted ray skims along the surface. That angle of incidence is the **critical angle**. Any bigger, and Snell's law would need $\sin\theta_2 > 1$, which is impossible. There is no refracted ray at all, and **all** the light reflects: **total internal reflection** (TIR).

DEFINITION

The **critical angle** C is the angle of incidence, in the medium with the higher refractive index, for which the angle of refraction is 90°.

Critical angle from Snell's law

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. At the critical angle, $\theta_1 = C$ and $\theta_2 = 90^\circ$. Put these into Snell's law:

$$n_1 \sin C = n_2 \sin 90^\circ$$

2. $\sin 90^\circ = 1$, so

$$\sin C = \frac{n_2}{n_1}$$

$\sin C$ can't be more than 1, so this only has an answer if $n_1 > n_2$. That's why TIR only happens going into a medium with a lower refractive index.

3. If the second medium is air, $n_2 = 1.00$. Calling the first medium's refractive index n:

$$\boxed{\sin C = \frac{1}{n}}$$

Critical angle (material to air)

FORMULA SHEET

$$\sin C = \frac{1}{n}$$

C: critical angle n: refractive index of the material for any other pair of media use $\sin C = n_2/n_1$

KEY IDEA

Predicting total internal reflection

TIR happens only if **both** conditions are met:

1. the light is travelling towards a medium with a **lower refractive index** (from slower to faster), and
2. the angle of incidence is **greater than the critical angle**.

If either fails, some light refracts out (and some is still partly reflected). In an exam, calculate C and compare it with the actual angle of incidence in words: "the angle of incidence, 50°, is greater than the critical angle, 41.8°, so total internal reflection occurs."

COMMON MISTAKE

$\sin C = 1/n$ is only for a boundary with air (or a vacuum). For glass (1.50) against water (1.33), $\sin C = 1.33/1.50 = 0.887$, so $C = 62.5^\circ$, not 41.8°. A ray hitting that boundary at 55° gets out into the water.

In an **optical fibre**, light in the glass core keeps meeting the cladding (slightly lower n) at more than the critical angle, so it is trapped and follows the fibre round bends. Diamonds sparkle because their critical angle is only 24.4°, so light inside is reflected many times before it escapes.

WORKED EXAMPLE

An optical fibre has a core of refractive index 1.50 and cladding of refractive index 1.45. (a) Calculate the critical angle at the core–cladding boundary. (b) Light enters the flat end of the fibre from air. Find the largest angle of incidence at the end for which the light is still totally internally reflected at the core–cladding boundary.

Show solution

(a) The cladding is not air, so use the general form:

$$\sin C = \frac{n_2}{n_1} = \frac{1.45}{1.50} = 0.967 \quad \Rightarrow \quad C = 75.2^\circ$$

(b) Let the ray refract at the end face at angle r to the axis of the fibre. The end face is at 90° to the side of the fibre, so the ray then hits the core–cladding boundary at $90^\circ - r$ to its normal. For TIR we need

$$90^\circ - r \geq 75.2^\circ \quad \Rightarrow \quad r \leq 14.8^\circ$$

Snell's law at the end face (air to core):

$$1.00 \times \sin\alpha = 1.50 \times \sin 14.8^\circ = 1.50 \times 0.256 = 0.384 \quad \Rightarrow \quad \alpha = 22.6^\circ$$

Any light entering at more than about 23° to the axis leaks out through the cladding.

Measuring the refractive index of a solid

PRACTICAL METHOD

Refractive index of a glass or Perspex block

1. Put the block on paper and draw round it, so you can replace it if it moves.
2. Aim a thin ray from a ray box (narrow slit) at the block. Mark the incident ray and the emerging ray with two crosses each, far apart.
3. Remove the block, join the entry and exit points to get the ray inside, and draw the normal at the point of entry.
4. Measure i and r with a protractor. Repeat for at least five values of i from about 10° to 70°.
5. Plot $\sin i$ **on the y-axis against** $\sin r$ **on the x-axis**. From $1.00 \times \sin i = n \sin r$, this is a straight line through the origin with **gradient** $= n$.

Reducing uncertainty: use a narrow ray and mark its centre; put the crosses far apart; use larger angles, where ±1° is a smaller percentage. Taking n from the gradient averages out random errors.

Another method: send the ray through the curved side of a semicircular block towards the centre of the flat face (it enters along a radius, so it doesn't bend). Increase the angle until the ray just grazes the flat face: that angle is C, and $n = 1/\sin C$.

EXAM TIP

A graph of i against r is a curve, so it can't give you n from a gradient. Always plot the **sines**.

Plane polarisation

Shake a rope up and down and the wave travels along it with the rope moving vertically. Shake it side to side and it moves horizontally. A transverse wave can oscillate in any direction **at right angles** to its direction of travel. Ordinary light from a bulb or the Sun is **unpolarised**: its oscillations point in every direction at right angles to the ray, changing randomly all the time. For light it is the **electric field** that oscillates (with a magnetic field at right angles to it).

DEFINITION

A **plane-polarised** wave is a transverse wave whose oscillations take place in **one plane only**, the plane that contains the direction the wave travels.

A **polarising filter** (a sheet of Polaroid) lets through only the part of the oscillation along its **transmission axis**. Think of the rope passing through a slot in a fence: vertical shakes get through a vertical slot, horizontal shakes are blocked. Unpolarised light through one filter comes out plane-polarised with half the intensity. A second filter, the **analyser**, passes all of it if its axis is parallel to the first and **none** if the axes are at 90° (**crossed**). Turn the analyser through 360° and you get two maxima and two minima.

KEY IDEA

Only **transverse** waves can be polarised. In a longitudinal wave such as sound the oscillation is along the direction of travel, so there is only one possible direction and nothing to filter. The fact that light can be polarised is evidence that light is a transverse wave.

Polarisation in use

- **Polarising sunglasses:** glare reflected off water and roads is partly polarised horizontally, so lenses with a vertical transmission axis cut it out.
- **LCD screens:** each pixel sits between polarising filters; a voltage on the liquid crystal controls how much light gets through.
- **Radio and microwaves:** transmitters send out plane-polarised waves, so a receiving aerial must be lined up the same way. Rotate a microwave receiver about the beam and the signal is smallest at 90° to the transmitter.
- **Stress analysis:** clear plastic between crossed polarisers shows coloured bands where it is stressed.

Diffraction and Huygens' construction

You can hear someone talking in the next room through an open door, even though you can't see them. Sound spreads out through the doorway; light, apparently, doesn't. Both are waves doing the same thing, and the difference is the **size of the gap compared with the wavelength**.

DEFINITION

Diffraction is the spreading out of a wave as it passes through a gap or around the edge of an obstacle.

KEY IDEA

Huygens' construction

Every point on a wavefront acts as a source of secondary circular **wavelets**, which spread out at the speed of the wave. A moment later, the new wavefront is the line that touches all the wavelets (their **envelope**).

On an open stretch of water, the wavelets from neighbouring points overlap and their envelope is just another straight wavefront, a wavelength further on. Now put a barrier with a gap in the way. Only the points in the gap send wavelets onwards.

- **Wide gap (much wider than λ):** in the middle the wavelets still build a straight wavefront; only at the edges is there nothing to balance them, so the wavefront curves just at its ends. The wave goes mostly straight on.
- **Gap about the same as λ:** the gap acts almost like a single point source, so the waves come out as near-circular arcs that fill the whole region behind the barrier.
- **Obstacle:** wavelets from the parts of the wavefront passing the edges spread sideways behind it. An obstacle about λ across barely leaves a shadow; a large one leaves a clear shadow.

KEY IDEA

- Diffraction is most noticeable when the **gap width is about the same as the wavelength**.
- Diffraction does **not change the wavelength, frequency or speed**: the wave is still in the same medium. Draw diffracted wavefronts with the same spacing as the incoming ones.
- If the gap is much smaller than λ, the wave still spreads in circles, but very little energy gets through.

This explains the doorway. Sound has wavelengths from centimetres to metres, similar to a doorway, so it diffracts strongly. Visible light (400–700 nm) is about a million times smaller than the gap, so it goes almost straight through. Long-wave radio (wavelength about 1 km) diffracts round hills into valleys that FM radio (about 3 m) can't reach. Diffraction is a signature of waves: electrons diffracting through crystals shows they behave as waves too (see *Particle Nature of Light*).

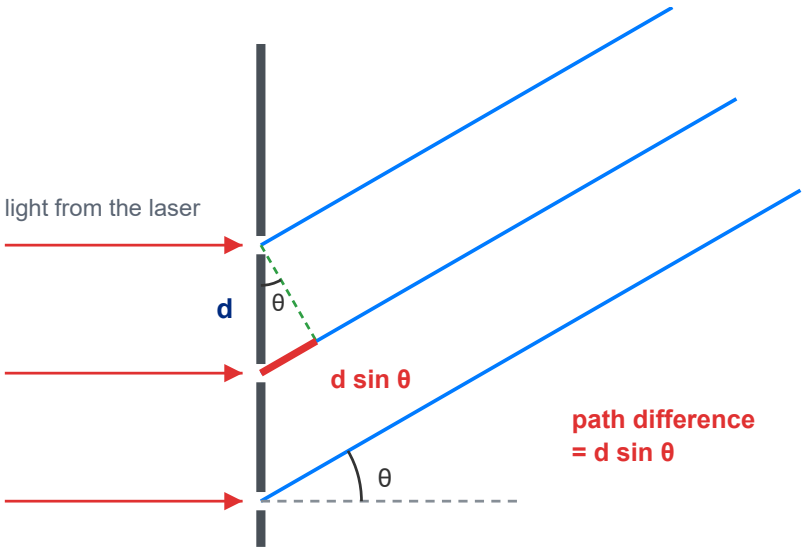
EXAM TIP

In "explain" questions, compare the gap or obstacle size with the **wavelength**; "the gap is small" earns nothing on its own. When sketching diffracted waves, keep the wavelength constant.

The diffraction grating

A **diffraction grating** is a slide with a huge number of narrow, equally spaced slits (lines), often hundreds per millimetre. Each slit diffracts light into a wide fan, and the light from all the slits overlaps and **interferes**. In most directions it cancels; only in a few sharp directions does it all arrive in phase, giving bright **maxima** called **orders**.

The **slit spacing** d is the distance from the centre of one slit to the centre of the next. A grating with N lines per metre has $d = 1/N$. For 300 lines per mm, $N = 3.00 \times 10^5$ per metre and $d = 3.33 \times 10^{-6}$ m.



Rays from neighbouring slits heading at angle θ to the normal. The ray from the lower slit travels an extra distance $d \sin \theta$ (red), the side of the right-angled triangle with hypotenuse d.

The diffraction grating equation

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Look at two neighbouring slits, a distance d apart, sending light at angle θ to the normal. The screen is so far away (compared with d) that the two rays are effectively parallel.

2. Drop a perpendicular from the top slit onto the lower ray. That makes a right-angled triangle with hypotenuse d, and the extra distance travelled by the lower ray is the side opposite the angle θ:

$$\text{path difference} = d\sin\theta$$

The angle at the top slit is θ because the perpendicular is turned 90° from the ray, just as the grating is turned 90° from the normal.

3. The waves arrive in phase, and interfere constructively, when the path difference is a whole number of wavelengths:

$$\boxed{n\lambda = d\sin\theta}, \qquad n = 0, 1, 2, \ldots$$

4. Every neighbouring pair has the same path difference, so in these directions light from **all** the slits is in phase. Move slightly away from that angle and the small phase differences add up across hundreds of slits until there is always a slit whose light cancels another's. That's why grating maxima are **bright and very sharp**.

Diffraction grating

FORMULA SHEET

$$n\lambda = d\sin\theta$$

n: order number (0, 1, 2, …) d: slit spacing / m θ: angle from the straight-through direction

How many orders can you see?

$\sin\theta$ can never be more than 1, so $n \leq d/\lambda$. Work out d/λ and round **down**. With 300 lines per mm and a 633 nm laser, $d/\lambda = 3.33 \times 10^{-6}/633 \times 10^{-9} = 5.27$, so the highest order is 5 and you see 11 spots (orders −5 to +5, including the zero order). More lines per mm means a smaller d, bigger angles, and fewer orders.

With **white light** the zero order is white (every wavelength goes straight through). In the other orders $\sin\theta \propto \lambda$, so each is a spectrum with **violet nearest the centre and red furthest out**, and higher orders overlap.

COMMON MISTAKE

Two common slips: (1) using the number of lines per mm as d. Convert to metres and take the reciprocal: 500 lines per mm gives $d = 1/(5.00 \times 10^5) = 2.00 \times 10^{-6}$ m. (2) Using the angle between the two first-order spots. θ is measured from the zero order, so it is **half** of that angle.

WORKED EXAMPLE

White light (wavelengths 400 nm to 700 nm) shines on a grating with 500 lines per mm. Show that the second-order and third-order spectra overlap, and explain why the third-order spectrum is incomplete.

Show solution

Slit spacing: $d = \dfrac{1}{5.00 \times 10^{5}} = 2.00 \times 10^{-6}\ \text{m}$.

The red end of the second order:

$$\sin\theta = \frac{2 \times 700 \times 10^{-9}}{2.00 \times 10^{-6}} = 0.700 \quad \Rightarrow \quad \theta = 44.4^{\circ}$$

The violet end of the third order:

$$\sin\theta = \frac{3 \times 400 \times 10^{-9}}{2.00 \times 10^{-6}} = 0.600 \quad \Rightarrow \quad \theta = 36.9^{\circ}$$

Third-order violet appears at a smaller angle than second-order red, so the two spectra overlap.

For third-order red, $\sin\theta = 3 \times 700 \times 10^{-9} / 2.00 \times 10^{-6} = 1.05$. That's more than 1, so no angle exists: the red end of the third-order spectrum is missing.

Core Practical 6: wavelength of laser light

CORE PRACTICAL

Determine the wavelength of light from a laser using a diffraction grating

Shine a laser through a grating of known lines per mm, with the grating at 90° to the beam, onto a screen or wall a distance D away (about 1 m). You'll see a row of bright spots.

1. Measure D from the grating to the screen with a metre rule or tape measure.
2. Measure the distance between the two first-order spots ($n = +1$ and $n = -1$) and halve it to get x_1. Measuring across both sides halves the uncertainty and means you don't have to judge exactly where the centre of the zero order is.
3. Calculate the angle: $\tan\theta = x/D$. Then $\lambda = \dfrac{d\sin\theta}{n}$.
4. Repeat for the higher orders. Either average the values of λ or plot $\sin\theta$ **against** n: a straight line through the origin with **gradient** $= \lambda/d$, so $\lambda = \text{gradient} \times d$.

Errors and improvements: measure between spot centres; use a large D so the percentage uncertainty in x is smaller (darken the room, as the spots get dimmer); check the grating is perpendicular to the beam (spots equally spaced either side of the centre); repeat and average.

Don't use the small-angle approximation $\sin\theta \approx \tan\theta$. Grating angles are often 10° to 50°, far too big for that.

Safety: never look into the beam or its reflections; keep the beam horizontal and below eye level; remove watches and rings; never point it at anyone. Even a low-power (class 2) laser can damage the retina if you stare into the beam.

WORKED EXAMPLE

In Core Practical 6, a student uses a grating with 300 lines per mm and a screen 1.20 m away. The two second-order spots are 0.985 m apart. (a) Calculate the wavelength of the laser light. (b) How many bright spots appear on a wide enough screen?

Show solution

(a) Distance from the centre to one second-order spot: $x_2 = 0.985/2 = 0.4925\ \text{m}$.

$$\tan\theta = \frac{x_2}{D} = \frac{0.4925}{1.20} = 0.410 \quad \Rightarrow \quad \theta = 22.3^\circ, \quad \sin\theta = 0.380$$

$$d = \frac{1}{3.00 \times 10^5} = 3.33 \times 10^{-6}\ \text{m}$$

$$\lambda = \frac{d\sin\theta}{n} = \frac{3.33 \times 10^{-6} \times 0.380}{2} = 6.33 \times 10^{-7}\ \text{m} = 633\ \text{nm}$$

(b) $d/\lambda = 3.33 \times 10^{-6}/6.33 \times 10^{-7} = 5.27$, so the highest order is 5. Counting $n = 0$ and orders 1 to 5 on each side: $2 \times 5 + 1 = 11$ spots.

EXAM TIP

Why a grating rather than two slits? Its maxima are **brighter**, **sharper** and **further apart** (tiny d), so their positions can be measured much more precisely.

Quick checks

QUICK CHECK

Light passes from water into air at an angle to the normal. Which of these stays the same?

- Its speed
- Its wavelength
- Its frequency
- Its direction

The frequency is set by the source; the boundary can't change the number of waves arriving each second. The speed increases, so the wavelength increases too ($v = f\lambda$), and the ray bends away from the normal.

QUICK CHECK

A ray inside glass ($n = 1.50$) hits a boundary with water ($n = 1.33$) at an angle of incidence of 55°. What happens?

- Total internal reflection, because 55° is greater than 41.8°
- Most of it refracts into the water, bending away from the normal
- It refracts into the water, bending towards the normal
- It passes into the water without changing direction

The critical angle here is $\sin^{-1}(1.33/1.50) = 62.5°$, not 41.8° (that's for glass to air). 55° is less than 62.5°, so the ray refracts out, at 67.5° to the normal, with a little partial reflection.

QUICK CHECK

Unpolarised light passes through two polarising filters. The second filter is rotated slowly through 360°. How many times does the transmitted intensity fall to zero?

- Never
- Once
- Twice
- Four times

The light is blocked whenever the two transmission axes are at 90° to each other, which happens at 90° and again at 270° of rotation.

QUICK CHECK

Water waves of wavelength 1.0 cm pass through a gap 5.0 cm wide. Which change would make them spread out the most?

- Increasing the wavelength to about 5 cm
- Decreasing the wavelength to 0.5 cm
- Widening the gap to 20 cm
- Doubling the amplitude of the waves

Diffraction is greatest when the gap width is about the same as the wavelength. Amplitude has nothing to do with how much a wave diffracts.

QUICK CHECK

Light of wavelength 600 nm falls on a grating with 400 lines per mm. What is the highest order of maximum that can be seen?

- 2
- 4
- 5
- 9

$d = 1/(4.00 \times 10^5) = 2.50 \times 10^{-6}$ m, so $d/\lambda = 4.17$. Round down: the highest order is 4. (9 is the total number of spots, orders −4 to +4.)

QUICK CHECK

Which observation shows that light is a transverse wave?

- Light can be refracted
- Light can be diffracted
- Light can be plane polarised
- Light can be totally internally reflected

Refraction, diffraction and reflection happen to longitudinal waves such as sound too. Only transverse waves can be polarised.